King Fahd University of Petroleum and Minerals
Department of Mathematics
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Stat 201 Major Exam II Term 231

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes Original course: STAT 319

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1. (Q4.129) The life (in hours) of a computer processing unit (CPU) is modeled by a Weibull distribution with parameters $\beta = 2$ and $\delta = 900$ hours. What is the probability that the CPU fails before the mean life of the CPU?

(a) 0.4559

(b) 0.5441

(c) 0.6099

(d) 0.3901

(e) 0.5000

2. (Q8.11) The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed with standard deviation 3. How large must n be if the length of the 99% CI is to be 1?

(a) 35

(b) 98

(c) 239

(d) 60

(e) 139

3. (Q8.18) Past experience has indicated that the breaking strength of yarn used in manufacturing drapery material is normally distributed with standard deviation of 2 psi. A random sample of nine specimens is tested, and the average breaking strength is found to be 98 psi. What is the length of a 95% confidence interval on the true mean breaking strength?

(a) 1.1

(b) 1.706

(c) 2.2

(d) 2.613

(e) 1.306

4. (Example 7.1) An electronics company manufactures resistors that have a mean resistance of 100 ohms and a standard deviation of 10 ohms. The distribution of resistance is normal. Find the probability that a random sample of size 25 resistors will have an average resistance of fewer than 95 ohms.

(a) 0.9938

(b) 0.2776

(c) 0.6914

(d) 0.0062

(e) 0.5000

5. (Q8.1-c) For a normal population with known variance, the confidence level for the interval $\bar{x} \pm 1.85\frac{\sigma}{\sqrt{n}}$ is

(a) 96.78%

(b) 93.56%

(c) 95%

(d) 98.39%

(e) 90%

6. (Q4.48) The probability density function of the weight of packages delivered by a post office is

$$f(x) = \frac{70}{69x^2}, \qquad 1 \leq x \leq 70$$

If the shipping cost is \$2.50 per pound, what is the average shipping cost of a package?

(a) \$10.469

(b) \$2.5

(c) \$6.81

(d) \$4.31

(e) \$10.775

7. (Example 8.8) In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. A 95% confidence interval estimate of the proportion of bearings in the population that exceeds the roughness specification is given by

(a) Between 4.92% and 18.61%

(b) Between 2.77% and 20.76%

(c) Between 6.02% and 17.51%

(d) Between 8.44% and 15.08%

(e) Between 7.29% and 16.24%

8. (Q8.9) Suppose that 100 random samples of water from a freshwater lake were taken and the calcium concentration (milligrams per liter) measured. A 95% CI on the mean calcium concentration is $0.49 \leq \mu \leq 0.82$. The length of a 99% CI calculated from the same sample data would be

(a) same length

(b) sometimes wider and sometimes shorter

(c) we need more information to tell

(d) shorter

(e) wider

9. (Q4.141) The length of time (in seconds) that a user views a page on a Web site before moving to another page is a lognormal random variable with parameters $\theta = 0.5$ and $\omega^2 = 2$. By what length of time have 50% of the users moved to another page?

(a) 1 second

(b) 6.782 seconds

(c) 0.147 seconds

(d) 2.495 seconds

(e) 1.649 seconds

10. A machine filling mini boxes of raisins fills the boxes so that the weight of the boxes has a population mean 14.1 grams and a population standard deviation 1.4 grams. A random sample of 49 boxes is selected. What is the probability that the total weight of the raisins will be at most 675 grams?

(a) 0.4090

(b) 0.9438

(c) 0.5

(d) 0.0526

(e) 0.5910

11. (Example 7.1 and Q7.9) An electronics company manufactures resistors that have a mean resistance of 100 ohms and standard deviation of 10 ohms. The distribution of resistance is normal. How large must the random sample be if you want the standard error of the sample average to be no more than 1.5 ohms?

(a) 45

(b) 7

(c) 1

(d) 350

(e) 35

12. (page 239) For a sample of size 30 selected randomly from a lognormal distribution with $\theta = 2$ and $\omega = 0.75$, the sampling distribution of the sample mean will be

(a) Lognormal with standard error 1.55

(b) Lognormal with standard error 8.51

(c) Normal with standard error 8.51

(d) Approximately normal with standard error 8.51

(e) Approximately normal with standard error 1.55

13. (Example 8.9) What is the needed sample size to estimate the true population proportion if we wanted to be at least 95% confident that our estimate of the true population proportion was within 0.05 regardless of the true value of the population proportion p?

(a) 164

(b) 271

(c) 256

(d) 385

(e) 666

14. (Example 4.2) Let the continuous random variable X denote the diameter of a hole drilled in a sheet metal component. The target diameter is 12.5 millimeters. Most random disturbances to the process result in larger diameters. Historical data show that the distribution of X can be modeled by a probability density function

$$f(x) = 20e^{-20(x-12.5)}, \qquad x \geq 12.5$$

If a part with a diameter greater than 12.60 mm is scrapped, what proportion of parts is scrapped?

(a) 0.86467

(b) 0.13533

(c) 0.5

(d) 0.99996

(e) 0.00004

15. (Example 7.2) Suppose that the random variable X has the continuous uniform distribution with probability density function given by

$$f(x) = 0.5, \qquad 4 \leq x \leq 6$$

How many observations should be selected in a sample from this distribution, if the sample mean is to be fewer than 5.2 with 99% assurance?

(a) 25

(b) 30

(c) 5

(d) 7

(e) 46

16. (Q4.62) The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce. If all cans less than 12.3 or more than 12.5 ounces are scrapped, what proportion of cans is scrapped?

(a) 0.8413

(b) 0

(c) 0.6826

(d) 0.1587

(e) 0.3174

17. (Q4.96) The life of automobile voltage regulators has an exponential distribution with a mean life of six years. You purchase a six-year-old automobile, with a working voltage regulator and plan to own it for six years. What is the probability that the voltage regulator fails during your ownership?

(a) 0.3678

(b) 0.4346

(c) 0.5654

(d) 0.6321

(e) 0.5

18. (Q7.16) Scientists at the Hopkins Memorial Forest in western Massachusetts have been collecting meteorological and environmental data in the forest data for more than 100 years. In the past few years, sulfate content in water samples from Birch Brook has averaged 7.48 mg/L with a standard deviation of 1.60 mg/L. What is the standard error of the sulfate in a collection of 10 water samples?

(a) 0.1834 mg/L

(b) 1.6 mg/L

(c) 31.623 mg/L

(d) 0.5059 mg/L

(e) 7.48 mg/L

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$\mu = 900(0.5)\sqrt{\pi} = 797.6$; $P = 1 - e^{-(0.8862)^2} = 0.5441$.
2	**(c)**	$E = 0.5$: $n = (2.575 \cdot 3/0.5)^2 = 238.7$, so $n = 239$.
3	**(d)**	Length $= 2(1.96)(2)/\sqrt{9} = 2.613$.
4	**(d)**	$\sigma_{\bar{X}} = 2$, $z = -2.5$, $P = 0.0062$.
5	**(b)**	$P(Z > 1.85) = 0.0322$, $\alpha = 0.0644$, level $= 93.56\%$.
6	**(e)**	$E(X) = \frac{70}{69} \ln 70 = 4.31$; cost $= 2.5(4.31) = \$10.775$.
7	**(a)**	$\hat{p} = 0.1176$; margin 0.0685; interval 4.92% to 18.61%.
8	**(e)**	Higher confidence means a larger critical value, so the interval is wider.
9	**(e)**	Median $= e^{\theta} = e^{0.5} = 1.649$ seconds.
10	**(d)**	$\bar{X} < 13.776$, $z = -1.62$, $P = 0.0526$.
11	**(a)**	$10/\sqrt{n} \le 1.5 \Rightarrow n \ge 44.44$, so $n = 45$.
12	**(e)**	CLT: approximately normal; $\sigma = 8.506$, so SE $= 8.506/\sqrt{30} = 1.55$.
13	**(d)**	$p = 0.5$: $n = 0.25(1.96/0.05)^2 = 384.16$, so $n = 385$.
14	**(b)**	$P(X > 12.6) = e^{-20(0.1)} = e^{-2} = 0.13533$.
15	**(e)**	$\mu = 5$, $\sigma^2 = 1/3$; $0.2\sqrt{3n} = 2.33$ gives $n = 45.2$, so $n = 46$.
16	**(e)**	$z = \pm 1$: $2P(Z < -1) = 2(0.1587) = 0.3174$.
17	**(d)**	Memoryless: $P(X < 6) = 1 - e^{-1} = 0.6321$.
18	**(d)**	$1.60/\sqrt{10} \approx 0.5059$ mg/L.